How-To Library

Informatica

Resuming PowerExchange Oracle LogMiner CDC after Unexpected DDL Changes to a Source

© Copyright Informatica LLC 2011, 2021. Informatica

Abstract

This article describes how to recover from a PowerExchange Logger for Linux, UNIX, and Windows failure that occurs because of unexpected DDL changes to an Oracle LogMiner CDC source table. With this method, you do not lose change data.

Supported Versions

- PowerExchange 10.0 or later

Table of Contents

Scenario. 2

Resuming CDC after an Unexpected DDL Change to a Source Table 2

Scenario

You administer the PowerExchange Oracle LogMiner change data capture (CDC) application in an environment that requires continuous capture of change data. Change activity occurs 24 x 7 on the Oracle source database, and you cannot stop it. Your goal is to keep CDC running and maintain data integrity on the target.

As recommended, your CDC implementation uses the PowerExchange Logger for Linux, UNIX, and Windows. Unexpectedly, the PowerExchange Logger abends, and the flow of change data to the target soon stops.

Based on PowerExchange message output, you determine that the PowerExchange Logger failure occurred because DDL changes were made to the Oracle source that you did not expect. For example, a column in a source table was dropped, a source table was dropped and then re-created, or the supplement log group for a source table was deleted. This type of situation can occur when an application that updates the Oracle database changes and the application change requires DDL changes to the Oracle source.

In this scenario, an application developer made an application change that required a column to be dropped from an Oracle source table. Neither the application developer nor the Oracle DBA advised you of this DDL change.

You now need to resume change data extraction processing as fast as possible to resynchronize the target table with the source table and maintain target data integrity. In your environment, loss of change data is not tolerated.

Resuming CDC after an Unexpected DDL Change to a Source Table

Complete the following steps to resume CDC processing after unexpected DDL changes with minimal or no change data loss. This procedure assumes that you cannot stop delete, insert, and update activity against the Oracle table.

1. Stop all PowerCenter workflows that extract change data for the source table that was affected by the DDL change.
2. Drop the supplemental log group for the source table if it was not already deleted.
3. In the PowerExchange Navigator, edit the capture registration for the source table to set the **Status** option to **History**. Then select the registration in the Resource Explorer and click the **Delete** button on the toolbar.

   You must delete the registration if you intend to re-create it under the same name.
4. To continue logging changes for other tables, warm start the PowerExchange Logger.

   A warm start uses the restart and sequence tokens from the last checkpoint file. You must have a recent checkpoint file. In the pwxccl command, set the coldstart parameter to N. For example:

   ```
   pwxccl coldstart=N config=pwx910/dbmover910.cfg cs=pwx910/pwxccl910.cfg &
   ```

5. In the PowerExchange Navigator, create a capture registration for the source table that reflects the DDL changes.

   Make sure you include the following settings:

   - Set the **Status** option to **Active**.
   - Select the **Execute DDL now** option to run the generated DDL that creates a supplement log group for the table when you click **Finish**. If you do not select this option, you must run the DDL manually before proceeding.
6. Change the target table definition to reflect the source table metadata changes, if necessary.
7. Get the current SCN for the Oracle instance. Use the following SQL query:

   ```
   SELECT DBMS_FLASHBACK.GET_SYSTEM_CHANGE_NUMBER FROM DUAL;
   ```
8. Use the SHUTDOWN or SHUTCOND command to stop the PowerExchange Logger, and then warm start the PowerExchange Logger.

   This step enables PowerExchange to start using the new capture registration.
9. In the PowerCenter Designer, import the extraction map for the altered source table. If the target table definition changed, also import the extraction map for the target table. Then edit the mapping if necessary.
10. Rematerialize the target table.

    You can use PowerExchange bulk data movement, PowerCenter, or another tool.
11. In the restart token file that is in the directory specified in the **RestartToken File Folder** attribute of the PWX Oracle CDC Real Time connection, enter a restart token value for the altered source table based on the Oracle current SCN from Step 7.

    If you do not know how to translate the Oracle current SCN into a restart token value, contact Informatica Global Customer Support.
12. Warm start the PowerCenter sessions that process the altered source table.

    Change data flow to the target resumes without change data loss. However, some transactions might be duplicated.

Author

Virginia Pfeifle

Acknowledgements

The author would like to acknowledge Dan Smith.